Cover Page

Concept Title	SMART AR: AI-Powered Markerless AR Learning
Theme	Quality Education / Emerging Technologies
Team Name	Team Spartan
Team ID	HSC|TN|02851
State / UT	Tamil Nadu

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Problem – Proposed Solution

CORE PROBLEM
- Textbooks and 2D diagrams can make abstract concepts difficult to visualize.
- Interactive learning tools may require special markers, costly resources, or dedicated apps.

PROPOSED SOLUTION
- Scan a textbook image using a web-based interface.
- Match the image against a curated library of learning images and 3D assets.
- When a match is found, display the linked 3D model in an AR view.

INNOVATION
- Combines image matching, reusable 3D assets and markerless AR in one learning workflow.
- Designed to be accessible through a web experience; unmatched images can show a clear not-found message.

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Technical Approach

TECH STACK (planned / as applicable)
- Front end: HTML, CSS, JavaScript
- 3D / AR: Three.js, WebXR (browser and device support dependent)
- Recognition: image matching / feature-based computer vision; Python or JavaScript implementation
- Assets: existing 3D model library with image-to-model mapping

WORKFLOW
1. Capture or upload a textbook image -> 2. Pre-process and compare with reference images
3. If a match is found, load its linked 3D asset -> 4. Render / place model in the AR view
5. If no match is found, ask the user to try another image

REPOSITORY: [Insert public GitHub/GitLab URL]
DEMO VIDEO: [Insert demo link or include in portal upload]
AI TOOL USAGE: Disclose the tools used and their role accurately before submission.

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Impact Potential

TARGET AUDIENCE
- College and school students; teachers and learning-content creators.

EXPECTED BENEFITS
- Makes difficult concepts easier to explore through interactive 3D visualization.
- Encourages curiosity, engagement and self-paced learning.
- Reuses existing 3D assets to reduce content-creation effort and support future expansion.

SDG ALIGNMENT
- SDG 4 ? Quality Education: supports engaging and accessible learning resources.
- SDG 9 ? Industry, Innovation and Infrastructure: applies emerging digital technologies.

STAKEHOLDERS & KPIs
- Students, educators, colleges and content providers. Track recognition accuracy, response time, successful model loads and user feedback.

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Feasibility and Risk Analysis

FEASIBILITY
- A web-based prototype can be built using common web technologies and a small curated asset library.
- Start with a limited set of textbook images and corresponding 3D models, then expand after testing.

KEY RISKS / CHALLENGES
- Similar-looking images, lighting, blur or camera angle may reduce matching accuracy.
- WebXR / AR support varies by browser and device.
- Large 3D assets may increase loading time; unmatched images may frustrate users.

MITIGATION
- Test with varied images and report measured accuracy; add confidence thresholds and a ?not found? path.
- Provide a non-AR 3D preview fallback, optimize asset sizes and test on target devices.
- Use only assets with appropriate licenses and collect user feedback during pilot testing.

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Team

TEAM COMPOSITION

- Harish S - Project coordination / web integration (confirm role)
- Jayaganesh U - AR / image-recognition workflow (confirm role)
- Harisankar R - 3D assets, testing and documentation (confirm role)

JOURNEY & CHALLENGES

- Building a practical learning experience around existing 3D assets rather than creating every model from scratch.
- Key challenge: reliable image matching and consistent AR support across devices.
- Next steps: complete the working prototype, test with students, measure results and refine the experience.

Note: Verify team members, registered team name and each person's role before final submission.

References

REFERENCES / TECHNICAL DOCUMENTATION

- Three.js documentation ? https://threejs.org/docs/
- MDN WebXR Device API ? https://developer.mozilla.org/docs/Web/API/WebXR_Device_API
- OpenCV documentation (image processing and feature matching) ? https://docs.opencv.org/
- United Nations Sustainable Development Goals ? https://sdgs.un.org/goals
- Project website ? https://smartar.netlify.app/
- Add any papers, datasets, model sources and licenses actually used in the implementation.